

FINAL PROJECT REPORT

Directional Traffic and Toll Revenue Forecasting for the Padma Bridge

Mawa- and Jajira-side forecasting, reconciliation and Eid direction analysis

CSE 4112: Machine Learning Laboratory
Group B2 | Batch 2021
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Ashique portion: directional forecasting update

Executive summary

This extension converts the bridge-wide forecasting system into a side-specific system for Mawa traffic, Jajira traffic, Mawa toll revenue and Jajira toll revenue. The cleaned daily source produced **1,492** complete modeling rows and **151** shifted, leakage-safe predictors. Persistence, seasonal persistence, Linear Regression, Random Forest, XGBoost, CatBoost and a residual MLP were evaluated chronologically.

XGBoost achieved the lowest direct test MAE for all four targets. Its MAPE was 8.88% for Mawa traffic, 8.85% for Jajira traffic, 8.34% for Mawa cash and 9.53% for Jajira cash. The total-and-share approach was also trained to guarantee arithmetic consistency when paired with a total forecast.

Main finding

Recent side history and nonlinear boosting provide useful direction-level forecasts. Total-and-share forecasting is operationally attractive because its two side outputs sum exactly to the supplied total, but its deployment accuracy depends on the quality of the separate total forecast.

Work completed

- Created side-specific lags, rolling means, rolling standard deviations, shares, directional differences and revenue-per-vehicle history.
- Trained 30 saved estimators across direct and total-and-share strategies.
- Evaluated on the final 224 days without random splitting.
- Produced model comparisons, date-level predictions and an Eid-relative side profile.

1. Introduction

The original system forecasts only bridge-wide total traffic and total toll revenue. Operational planning also benefits from knowing how demand is distributed between the Mawa and Jajira toll-plaza sides. Directional forecasts can support staffing, lane allocation and event-day traffic management, particularly when travel patterns reverse before and after Eid.

1.1 Objectives

- Predict Mawa and Jajira traffic separately.
- Predict Mawa and Jajira cash collection separately.
- Compare direct target models with total-and-share allocation models.
- Quantify side shares and ensure reconciled results sum to a bridge-wide total.
- Describe directional patterns around Eid without claiming unverified northbound/southbound labels.

1.2 Scope

The trained implementation is a one-day-ahead directional backtest using target-day calendar/weather information and bridge history ending at least one day earlier. The supplied brief also proposes 7- and 30-day directional forecasting; that joint extension requires the multi-horizon origin construction developed in the companion report and future weather assumptions.

2. Dataset and audit

Item	Value
Source	Padma daily traffic, toll, calendar and weather CSV
Raw coverage	26 Jun 2022 – 24 Aug 2026
Complete modeled rows	1,492
Predictors	151
Test period	2026-01-12 – 2026-08-24
Test observations	224

2.1 Targets and consistency

The four published side fields were cleaned from Indian comma grouping and retained as targets. Published totals were not replaced by sums of side values because the source contains a small number of inconsistencies. Direct forecasts may therefore have a non-zero reconciliation difference. In the total-and-share strategy, the predicted Mawa share is clipped to [0,1], the Jajira share is one minus the Mawa share, and allocation is made from the bridge total.

2.2 Leakage prevention

Every traffic, cash, share, difference and revenue-per-vehicle feature was shifted by at least one day. Rolling windows begin from the previous day. Same-day side observations never appear among predictors. Calendar, Eid and expected-weather values are the only target-date inputs.

3. Feature engineering and experimental design

Feature family	Examples
Calendar	Cyclic weekday, month and day-of-year; weekend; railway indicator
Events	Holiday, Eid, distance to Eid and signed Eid-relative day
Weather	Mean/max/min temperature, rainfall, humidity and wind speed
Side history	Lags 1, 2, 3, 7, 14 and 28; rolling mean and standard deviation
Directional structure	Mawa/Jajira share, directional difference and side revenue per vehicle
Bridge totals	Shifted total traffic and total cash histories

3.1 Chronological evaluation

Rows were ordered by date. The first 85% were used for model development and fitting and the final 15% were retained as the test period. This creates a realistic future evaluation and prevents later observations from influencing earlier predictions.

3.2 Models

Model	Purpose
Persistence / Seasonal	Last-day and last-week reference forecasts
Linear Regression	Interpretable linear baseline
Random Forest	Bagged nonlinear trees
XGBoost	Regularized gradient-boosted trees
CatBoost	Ordered boosting comparison for event/calendar interactions
Residual MLP	Neural model predicting change from the previous side value/share

4. Direct directional results

The table reports the best non-baseline direct model for each target. Cash values are in BDT. MAE is the primary selection measure; RMSE emphasizes large misses, MAPE gives a scale-free percentage and R-squared measures explained test variation.

Target	Best model	MAE	RMSE	MAPE	R-squared
jajira_cash	XGBoost	752,424	1,339,020	9.53%	0.732
jajira_traffic	XGBoost	701	1,444	8.85%	0.772
mawa_cash	XGBoost	731,438	1,370,969	8.34%	0.774
mawa_traffic	XGBoost	749	1,554	8.88%	0.801

Interpretation

XGBoost won all four direct targets on MAE. Traffic errors averaged roughly 700–750 vehicles per side, while cash errors averaged BDT 0.73–0.75 million per side. R-squared values from 0.732 to 0.801 indicate that the models explain a substantial share of daily test-period variation. The stronger Mawa traffic R-squared suggests somewhat more predictable variability on that side during this test window.

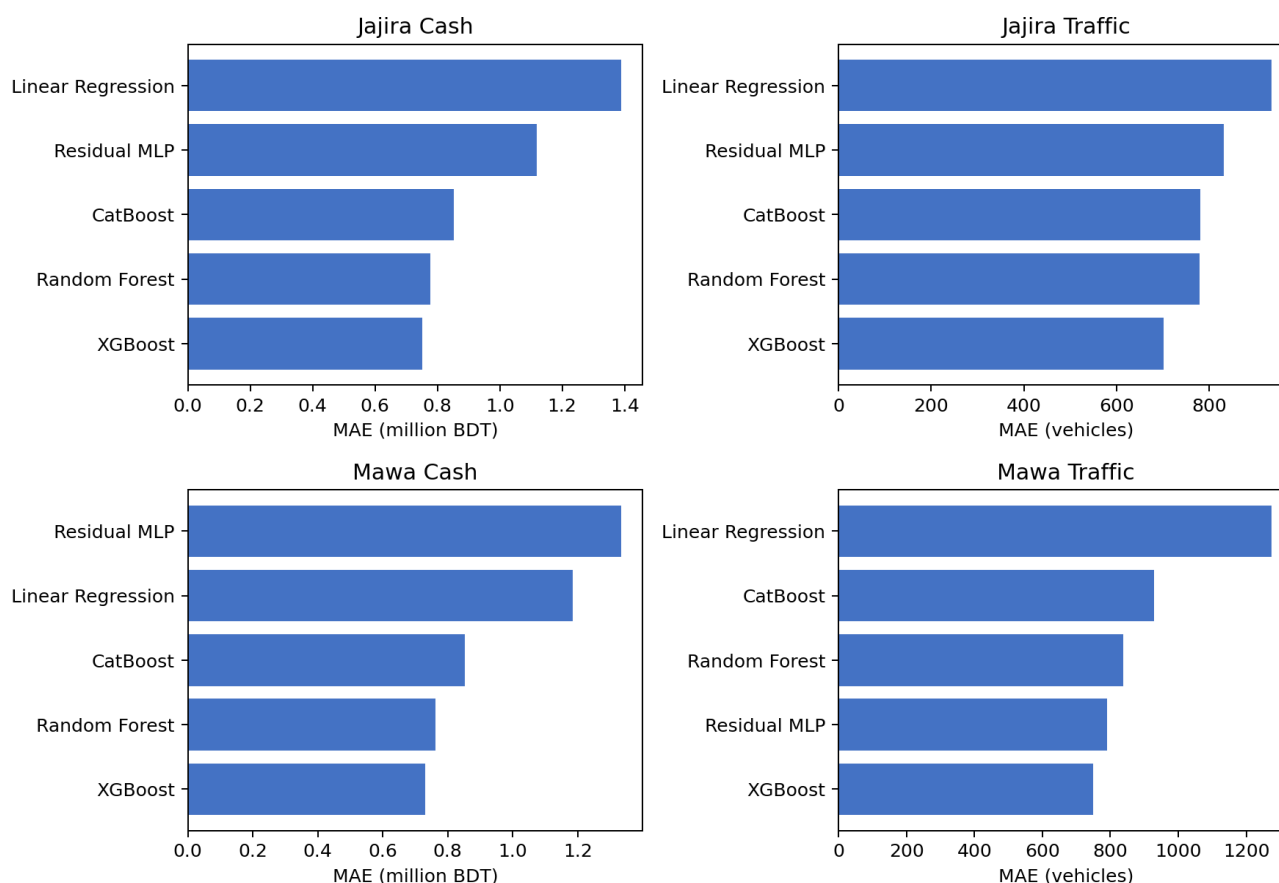


Figure 1. Test MAE comparison across trained direct model families. Cash panels use million BDT; traffic panels use vehicles.

5. Complete direct-model comparison

Target	Model	MAE	RMSE	MAPE	R ²
Jajira Cash	XGBoost	752,424	1,339,020	9.53%	0.732
Jajira Cash	Random Forest	776,669	1,491,874	9.90%	0.667
Jajira Cash	CatBoost	852,302	1,717,990	10.93%	0.558
Jajira Cash	Residual MLP	1,117,340	1,956,355	11.64%	0.427
Jajira Cash	Persistence-1	1,117,347	1,956,363	11.64%	0.427
Jajira Cash	Linear Regression	1,388,742	2,007,562	13.37%	0.397
Jajira Cash	Seasonal-7	1,813,947	3,182,942	17.44%	-0.516
Jajira Traffic	XGBoost	701	1,444	8.85%	0.772
Jajira Traffic	Random Forest	779	1,644	9.43%	0.704
Jajira Traffic	CatBoost	781	1,606	9.13%	0.717
Jajira Traffic	Residual MLP	832	1,515	9.50%	0.749
Jajira Traffic	Linear Regression	935	1,625	10.73%	0.711
Jajira Traffic	Persistence-1	1,166	2,047	11.63%	0.541
Jajira Traffic	Seasonal-7	1,967	3,709	19.33%	-0.507
Mawa Cash	XGBoost	731,438	1,370,969	8.34%	0.774
Mawa Cash	Random Forest	762,803	1,552,546	8.89%	0.710
Mawa Cash	CatBoost	853,495	1,827,093	9.31%	0.598
Mawa Cash	Linear Regression	1,184,518	1,781,422	10.13%	0.618
Mawa Cash	Residual MLP	1,333,274	2,259,028	13.90%	0.386
Mawa Cash	Persistence-1	1,333,283	2,259,041	13.90%	0.386
Mawa Cash	Seasonal-7	1,919,018	3,605,998	16.49%	-0.564
Mawa Traffic	XGBoost	749	1,554	8.88%	0.801
Mawa Traffic	Residual MLP	791	1,474	7.63%	0.821
Mawa Traffic	Random Forest	837	1,771	9.82%	0.741
Mawa Traffic	CatBoost	930	2,075	10.55%	0.644
Mawa Traffic	Linear Regression	1,275	2,094	12.35%	0.638
Mawa Traffic	Persistence-1	1,389	2,512	14.58%	0.479
Mawa Traffic	Seasonal-7	2,014	4,413	18.44%	-0.607

The persistence and seasonal baselines establish whether the machine-learning models add value beyond copying recent side values. Results differ by metric: for example, a residual model can have competitive percentage error while a boosted model better controls absolute error. Model choice should therefore consider both everyday MAE and high-demand RMSE, not a single accuracy percentage.

6. Direct versus total-and-share strategy

Direct models predict each side independently. Total-and-share models predict only the Mawa fraction; the Jajira fraction is its complement. This guarantees zero reconciliation error once the shares are multiplied by a common total forecast.

Criterion	Direct	Total-and-share
Side output	Four independent targets	Total × predicted share
Arithmetic consistency	Not guaranteed	Guaranteed
Dependency	Side model only	Share model plus total forecast
Interpretability	Individual levels	Demand level separated from allocation
Recommended use	Performance benchmark	Operational reconciled output

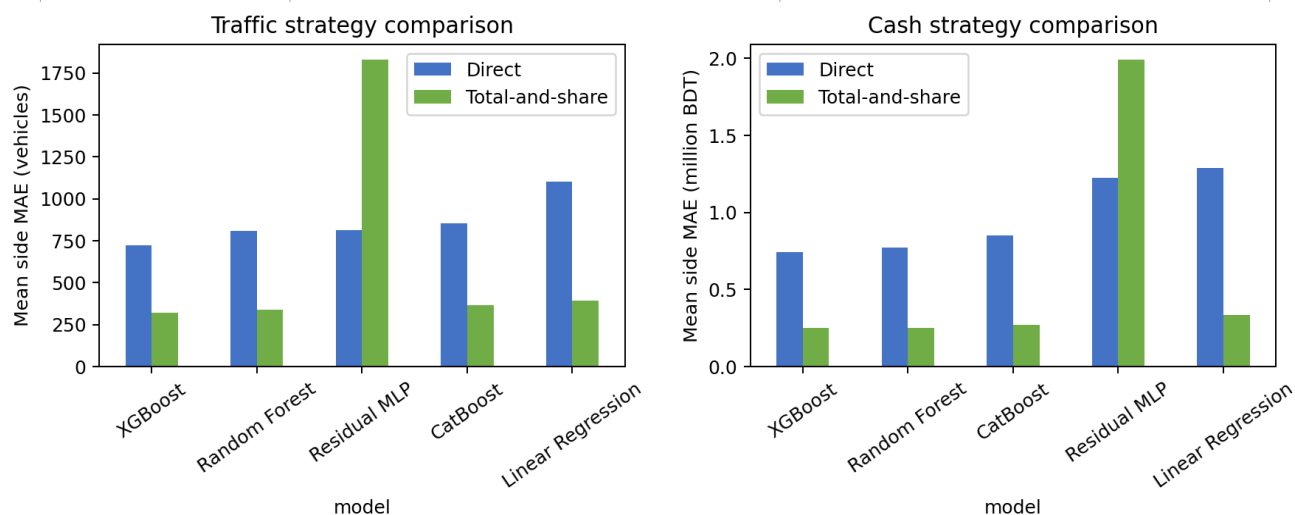


Figure 2. Average Mawa/Jajira MAE by strategy and model. Total-and-share values isolate allocation quality using the published total.

Important evaluation note

The total-and-share backtest used the published test-day total to isolate share-allocation accuracy. It is therefore an oracle-total comparison, not a complete deployable forecast. In production, the total must come from the separately trained total traffic/toll model, and its error will propagate into both sides.

7. Eid and holiday directional analysis

The signed Eid-relative profile groups observations from 14 days before through 14 days after Eid. The largest combined mean in this window occurred at relative day -1. This profile helps identify when a surge begins, whether Mawa or Jajira is dominant and when the side share returns toward its ordinary level.

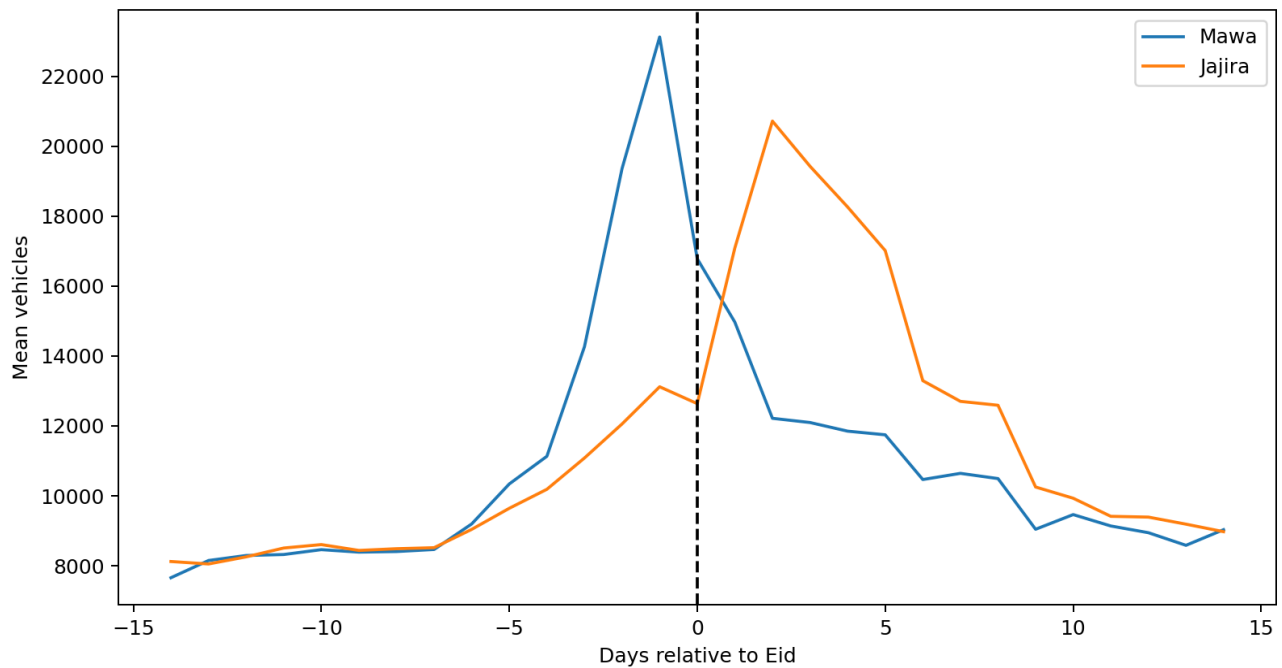


Figure 3. Mean Mawa and Jajira traffic by signed day relative to Eid. These are descriptive associations across the available Eid events.

The number of Eid events is small, so each relative-day mean is sensitive to individual years. The side labels should not be translated to northbound or southbound until Bangladesh Bridge Authority definitions confirm that interpretation.

8. Actual-versus-predicted behavior

The date-level plot compares each target with its selected direct prediction throughout the test period. Close tracking on routine dates supports operational usefulness; the widest gaps cluster around abrupt event-related changes, which are difficult to learn from only four years of history.

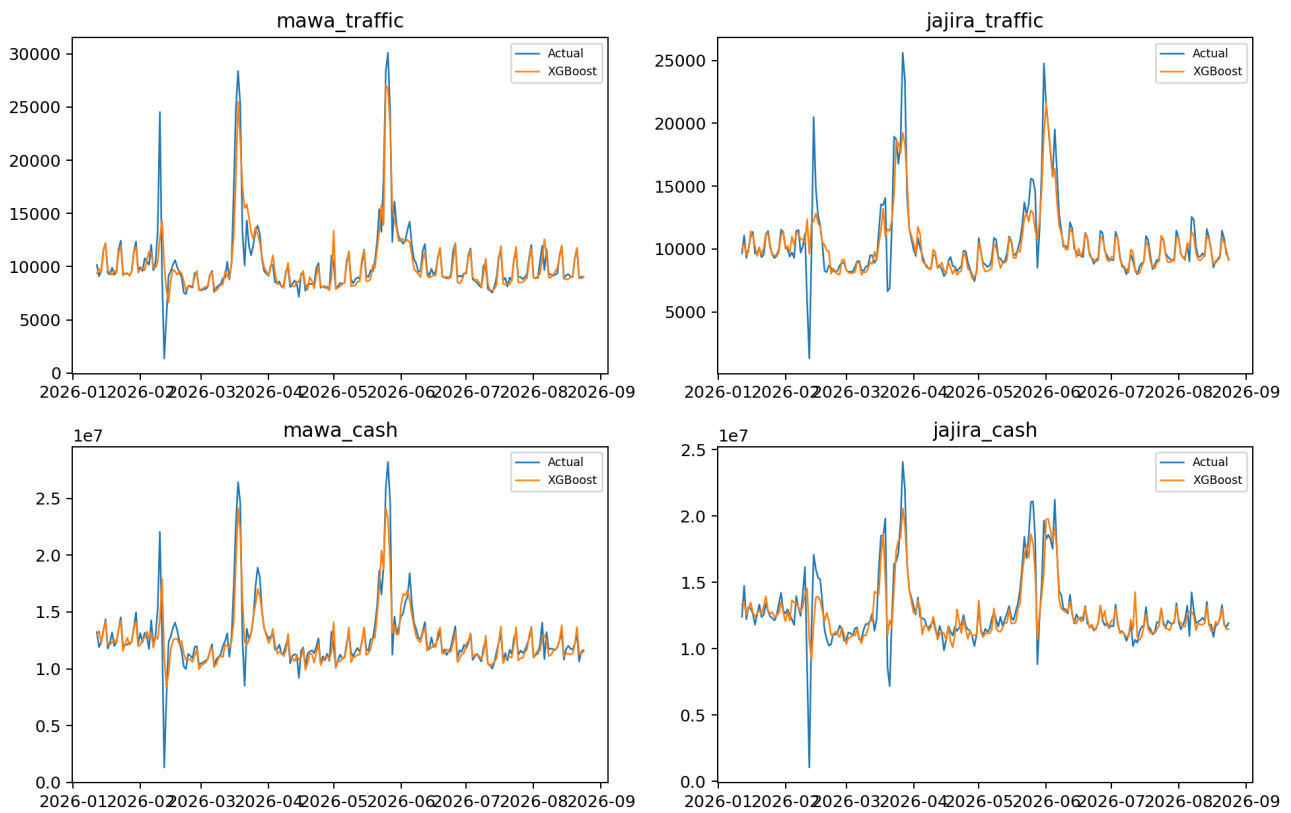


Figure 4. Actual side traffic/cash and the corresponding best direct model prediction during the held-out test period.

9. Limitations and responsible interpretation

- Only about four years of daily data and a small number of Eid events are available.
- Published side and total values occasionally disagree; the audit preserves rather than hides that issue.
- Observed historical weather is used for backtesting. Live use requires genuine forecast weather.
- Mawa and Jajira are safely described as toll-plaza sides, not confirmed travel directions.
- Unexpected closures, strikes, accidents and toll-rate changes are absent.
- Feature importance and event profiles are predictive/descriptive, not causal.

10. Conclusion

A complete directional forecasting extension was trained without altering the original project. XGBoost was the strongest direct model for every side target in the held-out test period. Total-and-share modeling provides exact reconciliation and a clear operational interpretation, provided that it is combined with an independently forecast bridge total. The main future priority is joint multi-horizon directional training with verified future calendars/weather and event-specific uncertainty intervals.

Appendix A. Delivered artifacts

The project folder contains 30 serialized estimators, complete direct/share metric tables, date-level direct predictions, an Eid profile, figures, a machine-readable run summary and the reproducible training program.

Appendix B. Reproduction

From the directional project folder, run: `python train_directional.py`. The script rebuilds features, fits all models and refreshes the outputs.

References

1. Bangladesh Bridge Authority, Padma Bridge traffic and toll records, <https://bba.gov.bd/>
2. Open-Meteo historical weather documentation, <https://open-meteo.com/en/docs/historical-weather-api>
3. CSE 4112 Group B2 directional forecasting brief, September 2026.